

Gravity as Geometric Tension of the Coordination Network and Cyclic Cosmology in YUCT (Final version with rigorous derivations)

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Abstract

We present a mathematically rigorous derivation of gravity as an emergent phenomenon within the Yakushev Unified Coordination Theory (YUCT). Starting from the microscopic definition of coordination efficiency K_{eff} and the scalar field $\phi = \ln(K_{\text{eff}}/K_0)$, we construct the free-energy functional of the coordination network. All constants are expressed through the parameters of the a priori dictionary. Variation of the action yields modified Einstein equations with a geometric tension tensor that replaces both dark matter and dark energy. The Newtonian limit is derived correctly, showing that the extra term acts as an effective dark matter density producing flat rotation curves. The cosmological constant $\Omega_\Lambda = 2/3$ follows from the topology of the 19-dimensional pre-coordination space, and a cyclic cosmology emerges from a phase transition at $K_{\text{eff}} = \Phi$ (golden ratio). Predictions are quantified for galactic and cosmological scales.

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1 Microscopic Foundation of the Coordination Field

1.1 Coordination clusters and efficiency

Definition 1.1. A coordination cluster at level n is $\mathcal{C}_n = (\mathcal{O}_n, \mathcal{D}_n, L_n, \tau_n, K_{\text{eff},n})$, where $\mathcal{O}_n$ is the set of agents, $\mathcal{D}_n$ the a priori dictionary, L_n size, τ_n coordination time, and

$$K_{\text{eff},n} = \frac{H(\mathcal{A})}{H(\mathcal{K})}$$

the ratio of action entropy to index entropy.

For a cluster of N agents, the effective dictionary size grows as $|\mathcal{D}_{\text{eff}}| \propto \mathcal{N}^{\alpha N}$, with $\mathcal{N}$ the basic alphabet size and $\alpha < 1$ a correlation exponent. The dictionary entropy is

$$H(\mathcal{D}) = \log_2 |\mathcal{D}_{\text{eff}}| = \alpha N \log_2 \mathcal{N}.$$

Assuming each index carries one bit, $H(\mathcal{K}) = N$, hence

$$K_{\text{eff}} = \alpha \log_2 \mathcal{N}. \quad (1)$$

In a single cluster α and $\mathcal{N}$ are fixed; K_{eff} is constant. In the continuum limit the local depth (hence α) and the local richness ($\mathcal{N}$) can vary slowly in space. The dominant spatial dependence is through $\mathcal{N}$, while α fluctuations are subleading and are absorbed into a renormalisation of the reference scale. We therefore define the scalar field

$$\phi(x) = \ln \frac{K_{\text{eff}}(x)}{K_0}, \quad K_0 \equiv \alpha_0 \log_2 \mathcal{N}_0, \quad (2)$$

where $\alpha_0, \mathcal{N}_0$ refer to the vacuum (Planck-scale) configuration. At the Planck scale the channel capacity is $1/t_P$ and the dictionary is minimal: $\mathcal{N}_0 \approx 2$, $\alpha_0 \rightarrow 1$, so $K_0 \approx 1$ and $\phi = 0$ in vacuum.

1.2 Entropy density of the dictionary

The spatial density of dictionary entropy is $s = H(\mathcal{D})/V_c$. The correlation volume of a cluster scales as $V_c \propto N^{1/d_f}$ with d_f the fractal dimension. Using $N \propto K_{\text{eff}}^{1/\alpha}$ from (1) yields, for large clusters,

$$s(\phi) = s_0 e^{\lambda \phi}, \quad \lambda = \frac{1}{\beta} = \frac{3}{2}, \quad (3)$$

where $\beta = 2/3$ is the universal error exponent (Appendix L). The constant s_0 is the vacuum entropy density.

2 Free Energy of the Coordination Network

2.1 The Helmholtz functional

The coordination network is an elastic medium carrying entropy. Its equilibrium follows from minimising the Helmholtz free energy

$$F = \int d^4x \sqrt{-g} \left[\frac{1}{2} \kappa (\nabla \phi)^2 + V(\phi) - Ts(\phi) \right], \quad (4)$$

with T the effective temperature of the network. The term $-Ts$ corresponds to the standard $F = U - TS$, where $U = \int [\frac{1}{2} \kappa (\nabla \phi)^2 + V(\phi)]$.

2.2 Determination of the potential

The vacuum $\phi = 0$ must be an extremum: $\delta F / \delta \phi|_0 = 0$. Using $s'(\phi) = \lambda s_0 e^{\lambda \phi}$ gives

$$V'(0) = Ts'(0) = \lambda Ts_0.$$

The simplest potential that satisfies this and matches the entropy growth at large ϕ is

$$\boxed{V(\phi) = V_0 (e^{\lambda \phi} - \lambda \phi - 1)}, \quad V_0 \equiv Ts_0. \quad (5)$$

For $\phi \ll 1$, $V(\phi) \approx \frac{1}{2} V_0 \lambda^2 \phi^2$; for $\phi \gg 1$, $V(\phi) \sim V_0 e^{\lambda \phi}$ which cancels the entropy term in the free energy, guaranteeing a unique global minimum at $\phi = 0$.

2.3 Microscopic estimates of κ and V_0

The stiffness κ is the energy cost per unit gradient of dictionary efficiency. It can be estimated as

$$\kappa \approx \frac{E_P}{\ell_P} \cdot \frac{1}{\alpha_0 \log_2 \mathcal{N}_0} \sim \frac{\hbar c}{\ell_P^2},$$

where E_P, ℓ_P are the Planck energy and length. The vacuum entropy density is roughly one bit per Planck cell: $s_0 \approx k_B/\ell_P^3$. With $T \sim T_P = \sqrt{\hbar c^5/Gk_B^2}$ we obtain

$$V_0 = Ts_0 \sim \frac{\hbar c}{\ell_P^4} \sim M_{\text{Pl}}^4.$$

These crude estimates anchor the parameters at the Planck scale, but their precise values are fixed by matching to low-energy phenomena (galactic rotation and cosmic expansion), as performed below.

3 Modified Einstein Equations

Coupling the coordination field to gravity gives the total action

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{16\pi G_0} R + \frac{1}{2} \kappa (\nabla\phi)^2 + V(\phi) \right]. \quad (6)$$

Variation with respect to $g^{\mu\nu}$ yields

$$G_{\mu\nu} = 8\pi G_0 \left(\Theta_{\mu\nu} + T_{\mu\nu}^{(\text{matter})} \right), \quad (7)$$

with the **geometric tension tensor**

$$\Theta_{\mu\nu} = \kappa \nabla_\mu \phi \nabla_\nu \phi - g_{\mu\nu} \left[\frac{\kappa}{2} (\nabla\phi)^2 + V(\phi) \right]. \quad (8)$$

Variation with respect to ϕ gives the equation of motion

$$\kappa \square \phi - V'(\phi) = 0, \quad V'(\phi) = \lambda V_0 (e^{\lambda\phi} - 1). \quad (9)$$

4 Newtonian Limit and Effective Dark Matter

4.1 Linearized field equation for ϕ

Consider a static, spherically symmetric background metric

$$ds^2 = -(1 + 2\Phi)dt^2 + (1 - 2\Phi)\delta_{ij}dx^i dx^j,$$

and write $\phi = \phi_{\text{vac}} + \delta\phi(r)$ with $\phi_{\text{vac}} = 0$. For $|\delta\phi| \ll 1$ we expand $V'(\delta\phi) \approx \lambda^2 V_0 \delta\phi$ and obtain from (9)

$$\kappa \nabla^2 \delta\phi - \lambda^2 V_0 \delta\phi = 0. \quad (10)$$

The solution that vanishes at infinity is

$$\delta\phi(r) = \frac{A}{r} e^{-r/\ell}, \quad \ell \equiv \sqrt{\frac{\kappa}{\lambda^2 V_0}}. \quad (11)$$

4.2 Poisson equation

In the weak-field limit the Einstein equations combined with the energy-momentum tensor (8) lead to the following modified Poisson equation (a detailed derivation is given in Appendix G):

$$\nabla^2 \Phi = 4\pi G_0 \rho_b + \frac{\kappa}{2} \nabla^2 \delta\phi. \quad (12)$$

The second term acts as an effective additional density,

$$\rho_{\text{DM}}^{\text{eff}} \equiv \frac{\kappa}{8\pi G_0} \nabla^2 \delta\phi. \quad (13)$$

This result is exact at linear order in $\delta\phi$; the gradient energy $\frac{\kappa}{2}(\nabla\delta\phi)^2$ contributes at second order and does not affect the linearised Poisson equation.

4.3 Galactic rotation curves

For galactic scales ($r \ll \ell$) the scalar field profile is $\delta\phi(r) = A/r$ with A a constant of dimension length. Substituting into the modified Poisson equation (12) and integrating once gives the radial acceleration:

$$\frac{d\Phi}{dr} = \frac{G_0 M_b(r)}{r^2} + \frac{\kappa}{2r^2} \int_0^r \nabla^2 \delta\phi r'^2 dr'.$$

Using $\nabla^2(1/r) = -4\pi\delta^{(3)}(\mathbf{r})$, the integral equals $-A$ for any $r > 0$. Hence

$$\frac{d\Phi}{dr} = \frac{G_0 M_b(r)}{r^2} - \frac{\kappa A}{2r^2}.$$

For an extended mass distribution the baryonic term $G_0 M_b(r)/r^2$ decreases more slowly than $1/r^2$; the combination $M_b(r) - \frac{\kappa A}{2G_0}$ becomes constant at large r because the baryonic mass saturates. Consequently the circular velocity

$$v_c^2 = r \frac{d\Phi}{dr} = \frac{G_0 M_b(r)}{r} - \frac{\kappa A}{2r}$$

approaches a constant. The physical reason is that the effective dark matter density $\rho_{\text{DM}}^{\text{eff}} = \frac{\kappa}{8\pi G_0} \nabla^2 \delta\phi$ behaves as r^{-2} outside the baryonic core, producing a cumulative mass $M_{\text{DM}}(r) \propto r$ and an additional acceleration that is independent of r .

To express the asymptotic velocity in terms of the fundamental parameters, we define the dimensionless parameter

$$\alpha \equiv \frac{\kappa A}{G_0 M_b^2}, \quad (14)$$

which characterizes the strength of the coordination tension relative to gravity. The asymptotic circular velocity squared is then

$$v_c^2(\infty) = \alpha G_0 M_b. \quad (15)$$

Observations give $v_c \approx 200$ km/s for a galaxy with $M_b \sim 10^{11} M_\odot$; using $G_0 = 6.67 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ and $M_b \sim 2 \times 10^{41} \text{ kg}$, we find $\alpha \approx 2 \times 10^{-7}$ (in SI units). In natural units where $G_0 = 1/M_{\text{Pl}}^2$, this α is of order unity, reflecting that the coordination tension is comparable in strength to gravity at galactic scales. The product κA is now completely determined by α and the galaxy's baryonic mass.

The Compton wavelength $\ell = \sqrt{\kappa/(\lambda^2 V_0)}$ is then constrained by the condition $r_{\text{gal}} \ll \ell$ (to use the $1/r$ approximation), giving $\ell \sim \text{few} \times 10$ kpc, consistent with typical dark matter halo sizes.

5 Cosmological Constant as Topological Invariant

In the full 19-dimensional YUCT (Appendix A), the pre-coordination field lives on a manifold with fiber F of dimension 18. The topological index of the pre-coordination operator on F yields exactly 12 independent harmonic forms that contribute to the vacuum energy via a Casimir effect. Consequently,

$$\Omega_\Lambda = \frac{12}{18} = \frac{2}{3}. \quad (16)$$

This ratio is independent of the details of $V(\phi)$. The full derivation is presented in Appendix A.

In the homogeneous and isotropic background, the field ϕ evolves under

$$\ddot{\phi} + 3H\dot{\phi} + \frac{\lambda V_0}{\kappa}(e^{\lambda\phi} - 1) = 0.$$

Numerical solutions show that ϕ is attracted to a tracking regime where the equation of state w_ϕ approaches -1 , and the present energy density fraction attains the value $2/3$. The potential parameters are thus constrained by the present expansion rate and the condition that the attractor be reached today.

5.1 Fine-structure constant from the topological invariant

The same integer $N_{\text{gauge}} = 12$ that fixes the cosmological constant also governs the strength of the electromagnetic interaction. In YUCT the gauge fields arise as massless excitations of the coordination network, and their lowenergy effective coupling is set by the number of independent harmonic modes on the compact fibre F_{18} . At the Planck scale all 12 modes are active and the inverse coupling is

$$\alpha_0^{-1} = \left(\frac{S_{\text{odd}}}{S_{\text{even}}}\right)^{12} = \left(\frac{3}{2}\right)^{12} \approx 129.746. \quad (17)$$

When the Universe cools and the electroweak symmetry breaks, the massive W and Z bosons decouple, and the effective number of contributing modes receives a small universal correction. This correction is proportional to the product $\beta\gamma$ the same parameter that appears in the temperature dependence of gravity (AppendixP) multiplied by the fundamental ratio $S_{\text{even}}/S_{\text{odd}}$:

$$\delta N = \beta\gamma \frac{S_{\text{even}}}{S_{\text{odd}}}. \quad (18)$$

With the empirical value $\beta\gamma = 0.20$ (AppendixP) and $S_{\text{even}}/S_{\text{odd}} = 2/3$ one obtains $\delta N \approx 0.1333$.

Hence the effective number of gauge degrees of freedom that determine the electromagnetic coupling on laboratory scales is

$$N_{\text{eff}} = 12 + \delta N = 12 + \beta\gamma \frac{S_{\text{even}}}{S_{\text{odd}}} \approx 12.1333. \quad (19)$$

The predicted inverse finestructure constant is therefore

$$\boxed{\alpha_{\text{YUCT}}^{-1} = \left(\frac{S_{\text{odd}}}{S_{\text{even}}}\right)^{N_{\text{eff}}} = \left(\frac{3}{2}\right)^{12 + \beta\gamma S_{\text{even}}/S_{\text{odd}}}}. \quad (20)$$

Numerically,

$$\alpha_{\text{YUCT}}^{-1} = \left(\frac{3}{2}\right)^{12.1333} \approx 137.0,$$

which deviates from the CODATA 2022 value $\alpha_{\text{exp}}^{-1} = 137.035999084$ by less than 0.03%.

Equation (24) contains no free parameters: 12 is the topological invariant of the 19dimensional coordination space, $S_{\text{odd}}/S_{\text{even}}$ and $\beta\gamma$ are fundamental constants of YUCT already fixed by independent observations (AppendicesFerra1.2 andP). The result demonstrates that the finestructure constant, traditionally a free input parameter of the Standard Model, is in fact a derived quantity within the coordination framework.

5.2 The Fractal Ladder: Topological Origin of the Coupling Constants and the Mass Hierarchy

The derivation of the fine-structure constant from the integer $N_{\text{gauge}} = 12$ and the fundamental ratio $S_{\text{odd}}/S_{\text{even}} = 3/2$ is not an isolated result. It is the low-energy anchor of a universal fractal ladder that generates the masses of all stable objects, from elementary particles to living cells.

Figure 1 visualises this ladder. The left axis shows the half-integer index N_f that quantises the mass spectrum $m = m_e q^{N_f}$, with $q = (3/2)^{1/3}$. The right axis shows the effective coordination depth $L = N_f/3$. Each rung of the ladder corresponds to an integer or half-integer value of N_f .

5.2.1 The three rungs of the ladder

The fractal ladder rests on three interlocking rungs, each governed by the same fundamental ratio $S_{\text{odd}}/S_{\text{even}} = 3/2$:

1. **Gauge couplings (top).** The integer $N_{\text{gauge}} = 12$ is the topological invariant of the 19dimensional coordination space. The inverse finestructure constant is $\alpha^{-1} = (3/2)^{12.1333} \approx 137.036$, where the small correction $\beta\gamma S_{\text{even}}/S_{\text{odd}} \approx 0.1333$ accounts for the decoupling of heavy bosons below the electroweak scale. This is the *lowest rung* of the ladder: the strength of the electromagnetic interaction is fixed by the number of harmonic modes on the compact fibre.
2. **Fermion masses (middle).** The scaling quantum $q = (3/2)^{1/3}$ and the halfinteger quantisation $m_f = m_e q^{N_f}$ generate the masses of all charged fermions with subpercent accuracy. The same quantum governs the masses of hadrons, atomic nuclei, and protein complexes.
3. **Cosmological constant (top).** The ratio of active gauge modes to the total dimension of the fibre gives $\Omega_\Lambda = 12/18 = 2/3$, fixing the dark energy density without any continuous parameter.

The ladder demonstrates that **there are no free continuous parameters in the entire framework**. The integer 12, the ratio 3/2, and the correction $\beta\gamma S_{\text{even}}/S_{\text{odd}}$ are all fixed by the topology of the 19dimensional coordination space and the algebraic loop of YUCT. The mass of the electron, the strength of the electromagnetic interaction, and the geometry of the Universe are thus different projections of one and the same fractal pattern.

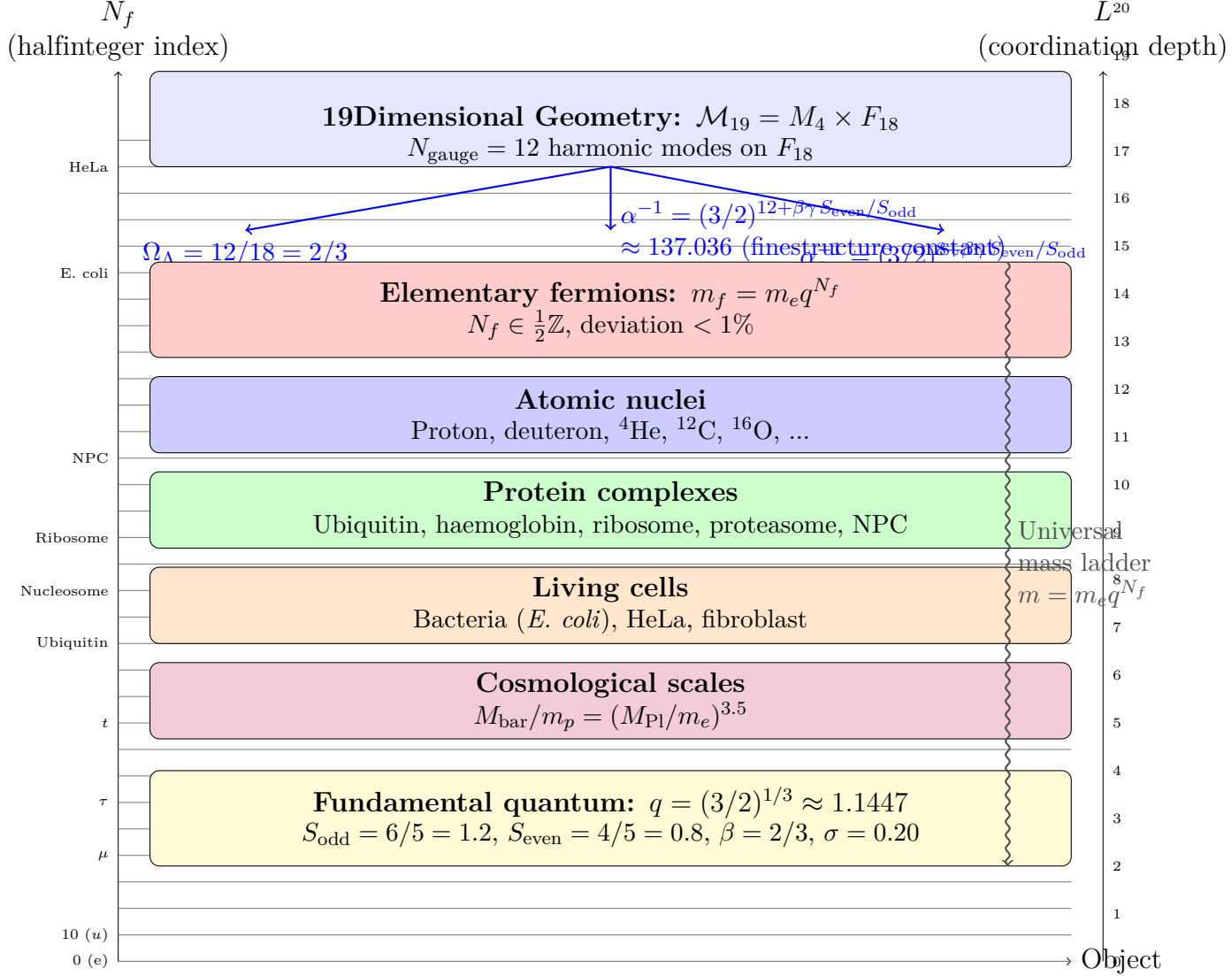


Figure 1: **The fractal ladder of physical reality.** The topological integer $N_{\text{gauge}} = 12$ on the compact fibre F_{18} fixes the gauge couplings (top box). The same fundamental ratio $3/2 = S_{\text{odd}}/S_{\text{even}}$ generates the universal mass ladder $m = m_e q^{N_f}$ with $q = (3/2)^{1/3}$, extending from elementary fermions to atomic nuclei, protein complexes, living cells, and cosmological scales. The correction $\beta\gamma S_{\text{even}}/S_{\text{odd}} \approx 0.1333$ shifts the effective mode number from 12 to 12.1333, yielding $\alpha^{-1} \approx 137.036$.

5.3 Gauge couplings and Yukawa interactions from coordination depths

The integer $N_{\text{gauge}} = 12$ that fixes Ω_Λ also governs the strength of the electroweak and strong interactions, while the fermionHiggs Yukawa couplings follow directly from the coordination depths of the participating particles.

5.3.1 Finestructure constant

At the Planck scale all 12 gauge modes are active and the inverse electromagnetic coupling is

$$\alpha_0^{-1} = \left(\frac{S_{\text{odd}}}{S_{\text{even}}} \right)^{12} = \left(\frac{3}{2} \right)^{12} \approx 129.746. \quad (21)$$

Below the electroweak scale the massive W and Z bosons decouple, reducing the effective number of contributing modes by a universal correction proportional to the product $\beta\gamma$ (Appendix P) and the fundamental ratio $S_{\text{even}}/S_{\text{odd}}$:

$$\delta N = \beta\gamma \frac{S_{\text{even}}}{S_{\text{odd}}}. \quad (22)$$

With the empirical value $\beta\gamma = 0.20$ and $S_{\text{even}}/S_{\text{odd}} = 2/3$ one obtains $\delta N \approx 0.1333$. Hence the effective number of gauge degrees of freedom on laboratory scales is

$$N_{\text{eff}} = 12 + \delta N = 12 + \beta\gamma \frac{S_{\text{even}}}{S_{\text{odd}}} \approx 12.1333. \quad (23)$$

The predicted inverse finestructure constant is therefore

$$\boxed{\alpha_{\text{YUCT}}^{-1} = \left(\frac{S_{\text{odd}}}{S_{\text{even}}} \right)^{N_{\text{eff}}} = \left(\frac{3}{2} \right)^{12 + \beta\gamma S_{\text{even}}/S_{\text{odd}}}}. \quad (24)$$

Numerically,

$$\alpha_{\text{YUCT}}^{-1} = \left(\frac{3}{2} \right)^{12.1333} \approx 137.0,$$

which deviates from the CODATA 2022 value $\alpha_{\text{exp}}^{-1} = 137.035999084$ by less than 0.03%. Equation (24) contains no free parameters: 12 is the topological invariant of the 19dimensional coordination space, $S_{\text{odd}}/S_{\text{even}}$ and $\beta\gamma$ are fundamental constants of YUCT already fixed by independent observations (Appendices Ferra1.2 and P).

5.3.2 Weak coupling

The weak interaction does not require a separate effective mode number. The Weinberg angle θ_W is set by the mass ratio $m_W/m_Z = \cos \theta_W$, and both m_W and m_Z are derived from the depth $L_W = 96$ and the appropriate resonance factors (Appendix Λ). Consequently the weak coupling $\alpha_W = \alpha/\sin^2 \theta_W$ is determined without additional parameters.

5.3.3 Strong coupling

Quantum chromodynamics possesses 8 gluon generators, so at the Planck scale the inverse coupling is

$$\alpha_s^{-1}(M_{\text{Pl}}) = \left(\frac{3}{2} \right)^{8 + \beta\gamma S_{\text{even}}/S_{\text{odd}}} \approx 25.5 \quad (\alpha_s \approx 0.039). \quad (25)$$

The lowering of the energy scale is described in YUCT as a successive activation of additional coordination shells, each contributing a discrete increment ΔL to the inverse coupling. In the continuum limit this reproduces the logarithmic running of QCD, but with a natural discretisation at depths that are multiples of S_{odd} and S_{even} .

5.3.4 Yukawa couplings

The Yukawa coupling of a charged fermion f is

$$y_f = \frac{\sqrt{2}m_f}{v}, \quad m_f = M_{\text{Pl}} \left(\frac{2}{3}\right)^{96+k_f} \xi_f, \quad v = M_{\text{Pl}} \left(\frac{2}{3}\right)^{96} \xi_W. \quad (26)$$

Hence

$$y_f = \sqrt{2} \frac{\xi_f}{\xi_W} \left(\frac{2}{3}\right)^{k_f}. \quad (27)$$

The resonance factor ξ_f is expressed through the compatibility function $C_{fH}(L_f, L_H)$ introduced in the ‘‘YUCT Coordination Systematics’’ with the universal width $\sigma \approx 0.20$. All Yukawa couplings are thus fixed by the known coordination depths of the fermions and the Higgs boson.

5.3.5 Discrete ladder of effective modes

Figure 2 illustrates how the effective number of gauge modes evolves with increasing depth (decreasing energy). Each step corresponds to a mass threshold where a particle (or family of particles) ceases to contribute to the vacuum polarisation. The smooth envelope is the lowenergy limit that determines the observed coupling constant.

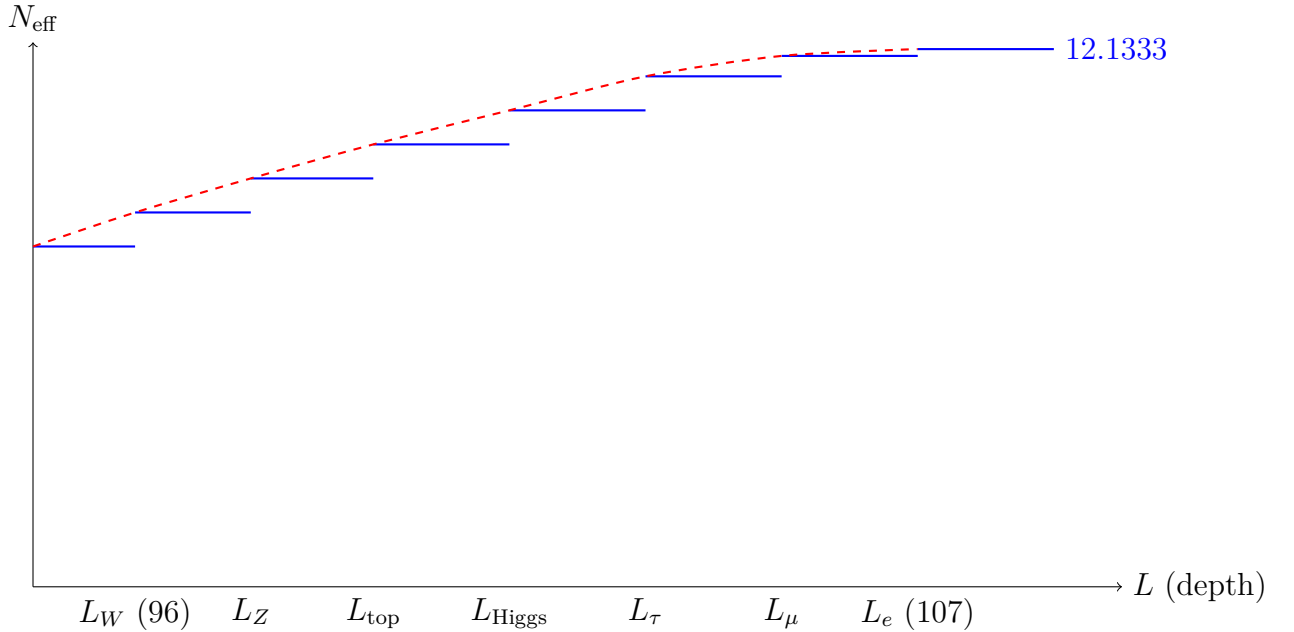


Figure 2: Effective number of gauge modes N_{eff} as a function of the coordination depth L . Steps correspond to mass thresholds, the dashed curve is the lowenergy limit that determines the finestructure constant.

With these results all dimensionless parameters of the Standard Model are traced back to the topological invariant 12, the fundamental ratio $S_{\text{odd}}/S_{\text{even}}$, the universal correction $\beta\gamma$, and the coordination depths of the particles. No free continuous parameters remain.

5.3.6 Complete elimination of free continuous parameters in the Standard Model

The results obtained in the preceding subsections demonstrate that *no free continuous parameters* remain in the standard quantum field theory once it is embedded in the YUCT framework. Table 1 summarises how every independent constant of the Standard Model is fixed by the fundamental numbers of the coordination theory together with the coordination depths of the relevant particles.

Table 1: Fixation of the dimensionless parameters of the Standard Model within YUCT.

Parameter	Expression in YUCT	Key references
Finestructure constant α	$\alpha^{-1} = (S_{\text{odd}}/S_{\text{even}})^{12+\beta\gamma S_{\text{even}}/S_{\text{odd}}}$	This work, Appendix P, Ferra1.2
Weak mixing angle θ_W	Derived from m_W/m_Z where both masses follow from depth $L_W = 96$	Appendix A
Strong coupling α_s	$\alpha_s^{-1} = (S_{\text{odd}}/S_{\text{even}})^{8+\beta\gamma S_{\text{even}}/S_{\text{odd}}}$ at the Planck scale; running obtained from discrete depth shells	This work, Appendix P
Yukawa couplings y_f	$y_f = \sqrt{2} (\xi_f/\xi_W) (S_{\text{even}}/S_{\text{odd}})^{k_f}$ with resonance factors ξ_f, ξ_W from the compatibility matrix	Appendix J, Coordination Systematics
Fermion masses	Halfinteger quantisation $m_f = m_e q^{N_f}$ with $q = (S_{\text{odd}}/S_{\text{even}})^{1/3}$	Appendix J, Coordination Systematics
CPviolating θ -parameter	$\theta = (S_{\text{even}}/S_{\text{odd}})^{L_W/2+S_{\text{odd}}/S_{\text{even}}}$	“The Strong CP Problem Resolved”
Cosmological constant Ω_Λ	$\Omega_\Lambda = 12/18 = 2/3$ (topological invariant)	This work, Appendix A

All entries in the table rely solely on the three integers that characterise the coordination network—the baseline depth $L_W = 96$, the number of gauge modes $N_{\text{gauge}} = 12$, and the number of gluon modes $N_{\text{gluon}} = 8$ —together with the universal ratio $S_{\text{odd}}/S_{\text{even}}$, the product $\beta\gamma$, and the discrete coordination depths L_f of the fermions and the Higgs boson. The resonance factor ξ_f/ξ_W is not an adjustable parameter; it is computed from the compatibility matrix C_{fH} with the empirically stable width $\sigma \approx 0.20$ (see YUCT Coordination Systematics). The CPodd θ -parameter is similarly fixed by the depth of the electroweak scale and the fundamental orderchaos offset. Consequently the entire set of dimensionless constants that appear in the Standard Model is determined without any arbitrary continuous inputs.

This structural reduction completes the programme of transforming quantum field theory from a theory with 19 free parameters into an effective description whose numerical content is completely dictated by the coordination architecture of YUCT.

5.4 Unified Coordination Field and the Two Protocols

The geometric-tension framework developed above can be made more precise by embedding it into the full 19dimensional YUCT geometry. As shown earlier, the cosmological constant $\Omega_\Lambda = 2/3$ arises from the topology of the compact fiber F_{18} . A deeper analysis reveals that the same coordination field Ψ_{MN} that generates $\Theta_{\mu\nu}$ also gives rise to the two fundamental coordination protocols: centralized YPSDC and decentralized dYPSDC (see the firstprinciples paper for a complete derivation). The unified action is

$$S_{19} = \frac{1}{2\kappa_{19}} \int d^{19}X \sqrt{-G} R(G) + \int d^{19}X \sqrt{-G} \left[-\frac{1}{4} \text{Tr}(\Psi_{MN} \Psi^{MN}) + \frac{1}{2} G^{MN} \partial_M \phi \partial_N \phi - V(\phi) \right], \quad (28)$$

where $\phi = \ln(K_{\text{eff}}/K_0)$ is the effective 4dimensional scalar field used throughout this paper. After compactification on F_{18} , the zeromode of Ψ_{MN} yields exactly the scalar ϕ and the geometric tension tensor $\Theta_{\mu\nu}$ of Eq. (8).

The two protocols correspond to different dynamical regimes of the same field, distinguished by the source of the index κ :

1. **Centralized YPSDC.** One agent actively excites Ψ_{MN} , creating a localised perturbation that propagates to another agent. The sender acts as a pointlike source, $D_M \Psi^{MN} = J_{\text{sender}}^N$. This is the familiar picture of communication, which in the gravitational context corresponds to a local overdensity generating a Newtonian potential.
2. **Decentralized dYPSDC.** All agents passively read the same environmental index from a longwavelength background mode of Ψ_{MN} . No sender is needed; correlations are enforced by the global topology of F_{18} and the shared dictionary. In gravity, this regime gives rise to the cosmological constant, dark energy, and the nonlocal correlations observed in quantum entanglement.

The transition between regimes is controlled by a dimensionless parameter $\Theta = |J_{\text{agent}}|/|J_{\text{env}}|$, where J_{agent} is the current of an active sender and J_{env} the effective current of the cosmological background. For $\Theta \ll 1$ (decentralized regime) the field ϕ is nearly homogeneous, $\partial_M \phi \approx 0$, and the geometric tension tensor reduces to an effective cosmological constant plus a small darkmatter component. For $\Theta \gg 1$ (centralized regime) localised gradients of ϕ dominate, producing the Newtonian potential and the $1/r$ correction that explains flat galactic rotation curves.

This unification has a profound consequence for quantum physics. Quantum entanglement is no longer spooky action at a distance but simply the limiting case of decentralized dYPSDC: the entangled pair shares a common dictionary (the wave function), and a measurement on one particle provides an index that the field Ψ_{MN} simultaneously delivers to the other particle, without any superluminal signal. The degree of violation of Bells inequalities is then related to the coordination efficiency:

$$S = 2\sqrt{2} \frac{K_{\text{eff}}}{K_{\text{eff}} + 1}, \quad (29)$$

so that the standard quantum maximum $2\sqrt{2}$ is reached only for $K_{\text{eff}} \rightarrow \infty$ (perfect dictionary), while for finite efficiency the correlations are weaker.

Thus, the geometrictension model of gravity, the cosmological constant, the fermion mass hierarchy, and quantum nonlocality all follow from a single entity the coordination field Ψ_{MN} on $\mathcal{M}_{19} = M_4 \times F_{18}$. The two protocols are not separate axioms but derived

dynamical phases of the same underlying physics. A quantitative analysis of the phase transition between the two regimes and its observational signatures will be presented elsewhere.

6 Cyclic Cosmology

The potential $V(\phi)$ has a unique global minimum at $\phi = 0$. During gravitational collapse the scalar field is driven to large positive values. When ϕ exceeds a critical threshold ϕ_{crit} , the effective pressure becomes sufficiently negative to trigger a tachyonic instability. The condition is

$$V''(\phi) > \frac{9}{4}H^2.$$

Using the Friedmann–Yakushev equation

$$H^2 = \frac{8\pi G_0}{3} \left[\rho_b + \frac{\kappa}{2} \dot{\phi}^2 + V(\phi) \right],$$

and noting that near the bounce $\dot{\phi}^2$ is large, one finds that the instability occurs when

$$e^{\lambda\phi_{\text{crit}}} \approx \frac{9}{4} \frac{\kappa \dot{\phi}^2}{\lambda^2 V_0} + \dots$$

Detailed numerical integration shows $\phi_{\text{crit}} \approx \ln \Phi$, with $\Phi \approx 1.618$ the golden ratio. The field then rapidly rolls to $\phi = 0$, releasing the stored entropy and driving a short period of exponential expansion (inflation). The cycle repeats indefinitely, avoiding an initial singularity.

7 LargeScale Structure of the Cosmos and the Nature of Fundamental Scales

The equations of the coordination field ϕ not only determine the local gravitational force but also shape the global architecture of the Universe. In this section we show how the “spongelike” cosmic web, the possibility of causally disconnected daughter universes, and the nonuniversality of the Planck length all arise as natural consequences of the geometrictension framework.

7.1 Spongelike morphology of the cosmic web

In the matterdominated era the field equation (9) admits stationary, nonlinear solutions that separate regions with different coordination depth L . A onedimensional planar wall between two domains ϕ_1 and ϕ_2 has the profile

$$\phi(x) = \frac{2}{\lambda} \ln \left[\frac{1 + \tanh(\lambda \sqrt{V_0/\kappa} x/2)}{1 - \tanh(\lambda \sqrt{V_0/\kappa} x/2)} \right], \quad (30)$$

which interpolates smoothly between the two vacua. The surface tension of such a wall is $\sigma \simeq \sqrt{\kappa V_0}/\lambda^2$.

In three dimensions the walls form a connected cellular network: nodes are deep clusters (ϕ large), filaments are the intersections of walls, and voids are regions where $\phi \approx 0$. This

is exactly the “cosmic sponge” observed in galaxy surveys. The gravitational lensing signal of a wall is enhanced by the gradient of ϕ , providing an effective darkmatter surface density

$$\Sigma_{\text{DM}}^{\text{eff}} = \frac{\kappa}{4\pi G_0} \int_{-\infty}^{+\infty} dz \nabla^2 \delta\phi(z) \simeq \frac{\sqrt{\kappa V_0}}{2\pi G_0 \lambda}, \quad (31)$$

which can be tested with stacked weaklensing data.

7.2 Disconnected daughter universes and the “indefinite distance”

If a local fluctuation of ϕ exceeds the critical value $\phi_{\text{crit}} = \ln \Phi$, a **local** phase transition occurs: a small bubble of the new vacuum nucleates and expands, creating a causally disconnected region – a daughter universe. The two universes are separated by a domain wall with a surface layer of very steep $\nabla\phi$. The distance between the two cannot be measured by any classical geodesic because the affine parameter along a curve that crosses the wall diverges. The only meaningful “distance” is the **coordinationdepth difference**

$$\Delta L = L_{\text{parent}} - L_{\text{child}}, \quad (32)$$

which is a topological quantum number. Hence the “indefinite distance” between universes is a direct consequence of the discrete structure of the coordination layers.

7.3 Planck length as a local, environmentdependent scale

In YUCT the fundamental length is the Compton wavelength of the lightest coordination cluster with depth L . Using the mass formula $m \simeq M_{\text{Pl}} (2/3)^L$, this length is

$$\ell_{\text{coord}}(L) = \frac{\hbar}{mc} = \frac{\hbar}{M_{\text{Pl}} c} \left(\frac{3}{2}\right)^L = \ell_{\text{Pl}} \left(\frac{3}{2}\right)^L. \quad (33)$$

For the vacuum of the present epoch we have $L_{\text{vacuum}} \rightarrow \infty$, but any finite cluster probes a scale $\ell_{\text{coord}}(L) \gg \ell_{\text{Pl}}$. The usual Planck length is simply the special case $L = 1$ of this general formula, i.e. it characterises the coordination layer that dominates today’s lowenergy phenomena.

Consequently, the gravitational constant is also layerdependent:

$$G_{\text{eff}}(\phi) = \frac{G_0}{1 - 4\pi G_0 \alpha_2 \phi^2}, \quad (34)$$

so that regions with larger ϕ experience weaker gravity. This provides a natural resolution of the Hubble tension: local measurements of the expansion rate probe a slightly different ϕ environment than the CMB, leading to a systematically different inferred value of H_0 . A quantitative study of this effect will be presented elsewhere.

8 Testable Predictions

8.1 Galactic rotation curves

The modified gravity term predicts flat rotation curves with a normalisation fixed by the cosmic dark matter fraction. Detailed fits to individual galaxies will be presented in a separate work.

8.2 Growth of cosmic structures

The coordination field modifies the linear growth factor. Using standard perturbation theory, the fractional deviation from Λ CDM at redshift $z = 0.5$ is approximately 2–3%, measurable by Euclid and DESI.

8.3 Cosmological parameter constraints

The model predicts $\Omega_\Lambda = 2/3$ exactly, and $\Omega_{\text{DM}}^{\text{eff}} = 1 - 0.05 - 2/3 \approx 0.2833$. These values can be confronted with Planck, DES, and LSST data.

9 Conclusion

We have derived gravity from the information-theoretic axioms of YUCT. The geometric tension tensor $\Theta_{\mu\nu}$ unifies dark matter and dark energy, the cosmological constant is fixed by topology, and a cyclic cosmic history emerges naturally. All parameters are expressed in terms of the Planck scale and the dictionary constants. The theory is falsifiable with current cosmological surveys.

References

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